

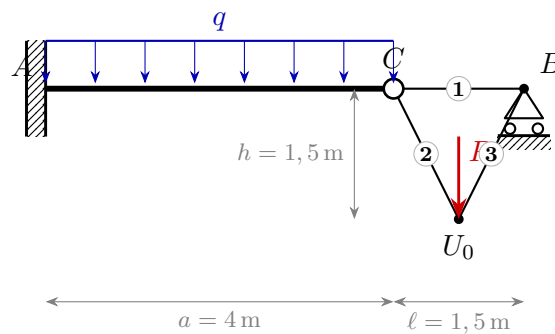
University of Natural Resources and Life Sciences, Vienna (BOKU)

VU Mechanik – LAWI 100515 (PI)

Exam **Group A**

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Model solution

**1. Static determinacy**Counting criterion $n = r - (3 + v)$ with r support reactions and v hinge conditions:

$$n = r - (3 + v) = 4 - (3 + 1) = 0 \quad \Rightarrow \quad \text{the system is statically determinate.}$$

2. Support reactions and hinge force

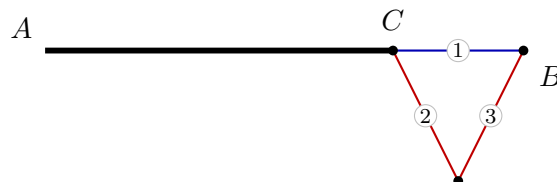
$$A_x = 0 \text{ kN}, \quad A_y = 30 \text{ kN}, \quad M_A = 80 \text{ kNm}, \quad B_x = 0 \text{ kN}, \quad B_y = 10 \text{ kN}$$

$$C_x = 0 \text{ kN}, \quad C_y = -10 \text{ kN}$$

3. Truss member forces

Member	Member force S [kN]	Type
1 (C–O1)	-5	compression
2 (C–U0)	11.18	tension
3 (U0–O1)	11.18	tension

red: tension (+) blue: compression (–)

**4. Section-force diagrams of the bending beam A–C**